PRD Genie - Metrics Workbook

Purpose :Defines per-agent and end-to-end metrics, LLM-as-a-judge rubrics, ground truth and the experiment evaluate PRD Genie in Langfuse.

Sources :PRD Genie problem statement (baseline T1-T12, capabilities), resource files (transcripts, brief, stake PRD template), Capstone Project Playbook (Steps 5-7 metrics, cost template), Epic/Story Upload Template.

Metric Definitions:1-2 metrics per agent (A1-A5), one for the code router, plus 2 end-to-end metrics. 'Source' says v metric comes from the Playbook or is additional.

Judge Prompts: Rubric prompts for Langfuse LLM-as-a-judge evaluators (judge model gpt-4o). Variables {{inpu {{expected_output}} map to the Langfuse dataset item and the trace output.

Ground Truth: Load as Langfuse dataset 'prdgenie-baseline-v1'. 'Expected output' column = expected_output cases (system must refuse to invent / must stop) are marked Y.

Cost Model Yellow cells are inputs (token estimates, rate, volume). All totals are formulas. Rate is the Playbook's 4o-mini example; re-check provider pricing before go-live

Scoring All judge scores are 0.0-1.0. A test passes when every applicable metric meets its pass threshold AN hallucination is detected.

Observation: Prompt Engineering had more gains than Model swap

| Workflow | Total score | Hallucination rate | Avg cost / run | Total cost |
|---|---|---|---|---|
| CHATGPT4 mini (v1) | 0.705 | 47% (17/36) | $0.0074 | $0.26 |
| CHATGPT4 mini + PROMPT (v4) | **0.779** | **45%** (14/31) | $0.0127 | $0.39 |
| Upgrade 5 (v1) | 0.695 | 58% (21/36) | **$0.0022** | $0.08 |

**What stands out:**

- **PROMPT v4 has the best total score and the fewest hallucinations.** It wins on M9 (0.91 vs about 0.76) and edges ahead on M2, M3 and M4. It is also the most expensive workflow, about 6× Upgrade 5 per run.
- **Upgrade 5 is the weakest( Not Prompt Engineered).** It has the most hallucinations and the lowest M3, M4 and M6. Its one clear win is M8 (0.79).

Note: there is an error in capturing cost and it is captured as Zero for LLM calls Hence not considering cost for comparisons

- **CHATGPT4 mini v1 leads on M5 and M6** (M6 0.59 vs 0.40 and 0.30).
- **PROMPT v4's lower M1 (0.65) is mostly a scoring difference, not a quality drop.** That workflow also scored M1 on the short-circuited "INSUFFICIENT" cases (31 scored runs vs 20 for the others), and several of those got 0. On full-pipeline runs only, its M1 is 0.71 vs 0.79 and 0.76.
- **CHATGPT4 mini's cost is mixed.** Its first batch of runs cost about $0.014 each, but runs 315–330 cost about $0.0015, roughly what Upgrade 5 costs. You may want to check whether the model changed between batches.